Large Language Models

Artificial Intelligence in Interactive Systems – 2025-07-10
Thomas Weber

Large Language Models

APPLE / TECH / ARTIFICIAL INTELLIGENCE

Apple is reportedly spending 'millions of dollars a day' training AI / Th
reportedly believes that Ajax,
language model, is more powe

By Monica Chin, a senior reviewer covering laptops and other gadgets.
Tom's Guide and Business Insider before joining The Verge in 2020.

TC

AI

Mistral AI makes its first large language model free for everyone

Devin Coldewey / 11:41 AM PDT • September 27, 2023

Comment

ARTIFICIAL INTELLIGENCE / TECH

Stability AI announces new open-source large language model / Stability AI, the same company behind the AI image generator Stable Diffusion, is now open-sourcing its language model, StableLM.

By Emma Roth, a news writer who covers the streami
and much more. Previously, she was a writer and edito

WILL KNIGHT BUSINESS APR 25, 2024 12:00 PM

**Meta's Open Source Llama 3 Is Alread
OpenAI's He**

Meta's decision to
models of OpenAI

nature

NEWS FEATURE | 13 December 2023

ChatGPT and science: the AI system was a force in 2023 – for good and bad

The poster child for generative AI software is a startling human mimic. It represents a potential new era in research, but brings risks.

By Richard Van Noorden & Richard Webb

STEPHEN ORNES SCIENCE MAR 24, 2024 8:00 AM

**Large Language Models' Emergent Ab
Mirage**

A new study suggests that sudden jumps in LLMs' abilities are
unpredictable, but are actually the consequence of how we m

TC

AI

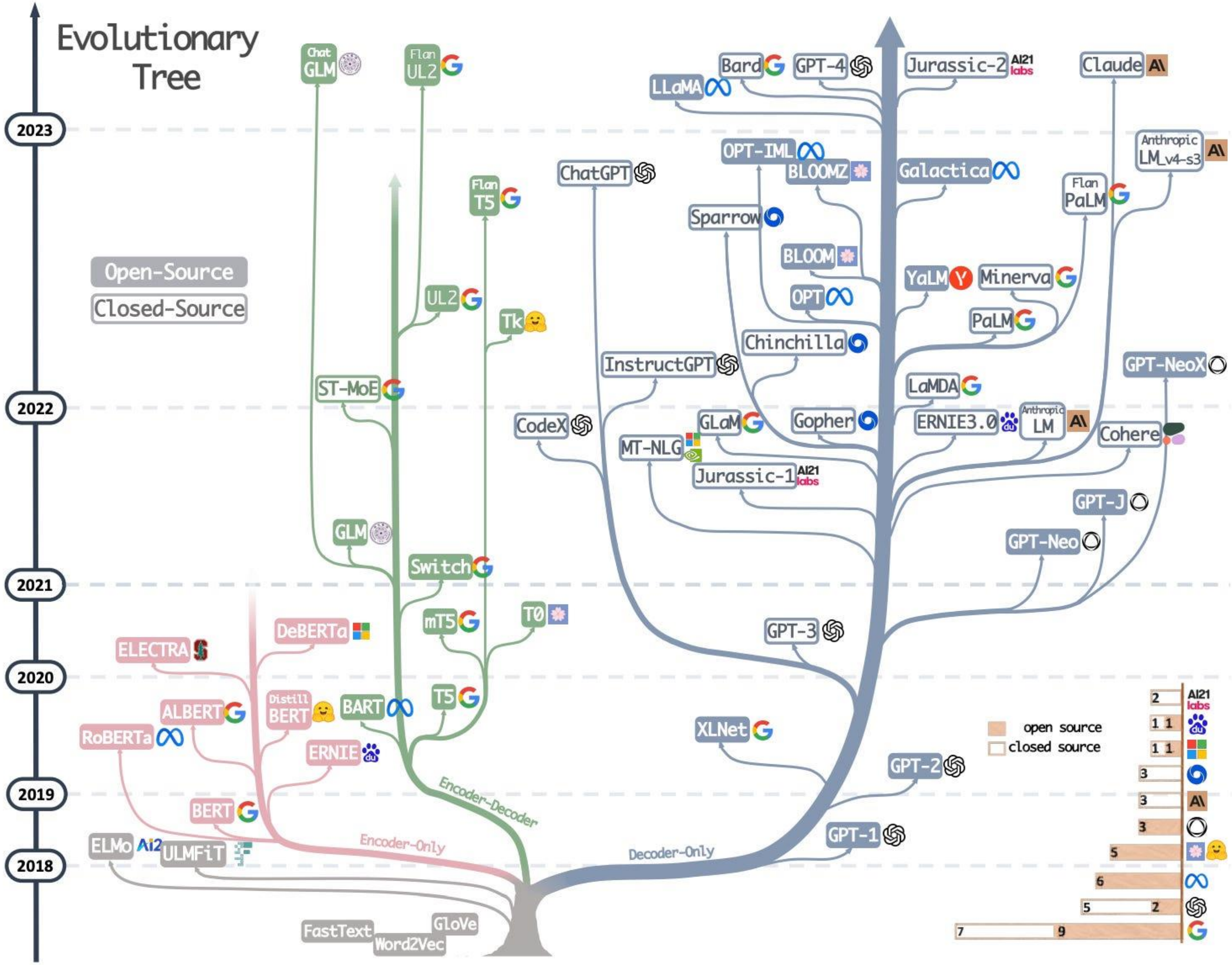
Apple brings ChatGPT to its apps, including Siri

Kyle Wiggers / 11:38 AM PDT • June 10, 2024 Comment

Image Credits: Apple

Apple is bringing ChatGPT, OpenAI's AI-
powered chatbot experience, to Siri and other

Large Language Models



https://www.interconnects.ai/p/llm-development-paths

Large Language Models

- ML models that generate natural language
- Wide array of use cases
- Easy to use via prompting

University library of the LMU Munich

central service facility of the Ludwig Maximilians University

The University Library (UB) of the Ludwig Maximilians University (LMU) (until 2012: Munich University Library) provides LMU scientists and students with literature and scientific information in printed and electronic form. In addition, the UB preserves and makes accessible the old holdings of valuable manuscripts and prints as the LMU's cultural heritage and offers open access platforms for electronic publications .

Table of contents

University library of the LMU Munich	
	
founding	1473
Duration	approx. 5.3 million media (as of December 2021)
Library type	University library
Location	Munich

Story

One year after the University of Ingolstadt was founded , the library of the artist faculty was built in 1473. In 1573, the vice-chancellor of the university, Martin Eisengrein , initiated the founding of an additional university library, which would also allow the university's secular professors free access to the books. The basis was 6,062 volumes by the Augsburg Bishop Johann Eglof von Knöringen , so that it was called the Knöringen University Library. The library of the Jesuit College in Ingolstadt was transferred to the university library when the order was dissolved in 1773.

In 1799 the university moved out of Ingolstadt. During the Landshut period from 1800 to 1826, the library was housed in the rooms of the expropriated Dominican monastery . During the secularization in Bavaria there was a significant increase in books. Through the "Decretum electorale" of 1803, the university library was entitled to take over the volumes of the Bavarian monastery libraries after the Munich Court Library . She also completely took over the holdings of the Seligenthal monastery and the Franciscan and

5 million books

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Library of Congress

- 32 millionen Books
- roughly 20 Terabyte

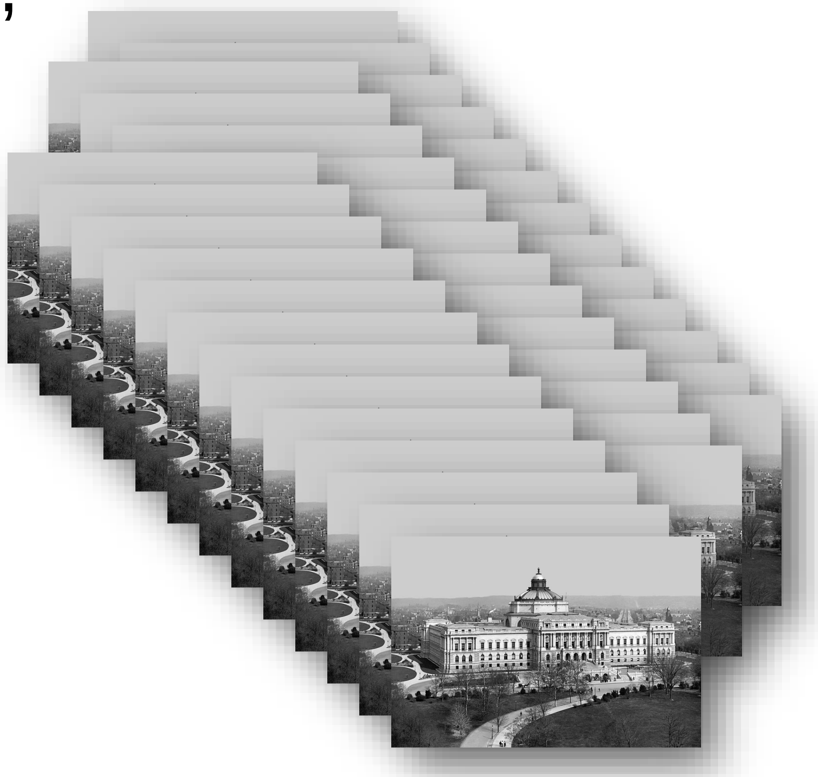
- 1 Terabyte
  = 1.000 GB
  = 1.000.000 MB

GPT-4

https://openai.com/research/gpt-4

- Generative Pretrained Transformer
- Multi Language
- Books, Online Platforms, Software Code, Articles, Entertainment, YouTube, E-Mails, …
- Training Data: ~1 Petabyte
  (~64 Mio. USD)

- 1 Petabyte ~ 50 libraries
  = 1.000 TB
  = 1.000.000 GB



Large Language Models

How do they work?

It

Large Language Models

How do they work?

It	is	31%
	can	27%
	has	24%
	will	18%

Large Language Models

How do they work?

It may

Large Language Models

How do they work?

It may

be	32%
not	32%
appear	24%
rain	12%

Large Language Models

How do they work?

It may seem

Large Language Models

How do they work?

It may seem like

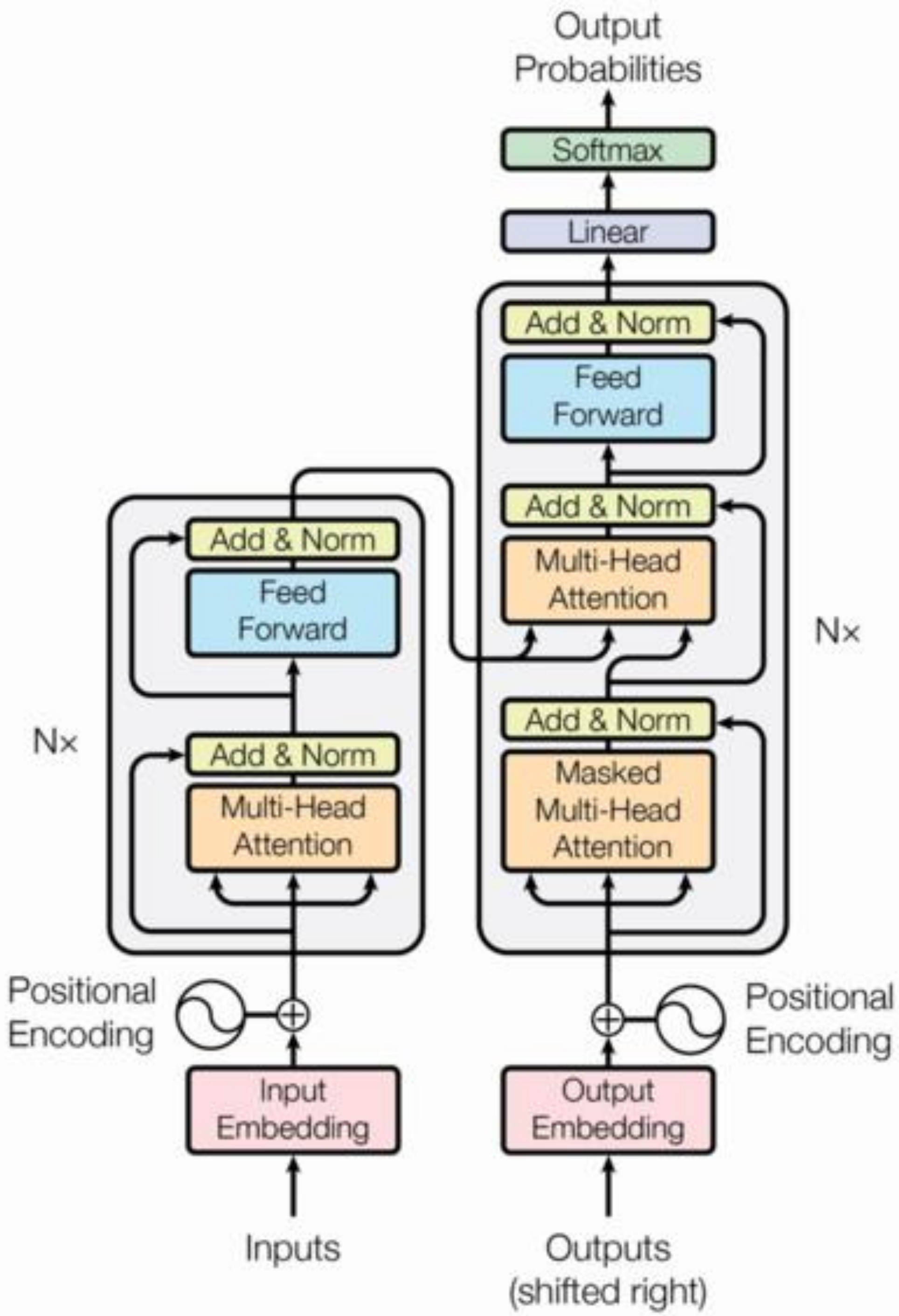
Large Language Models

How do they work?

It may seem like magic

Transformer Architecture

- Enabled by the **transformer architecture**
  - Predicts sequences, in this case of words
- Trained on massive amounts of data



https://dugas.ch/artificial_curiosity/GPT_architecture.html

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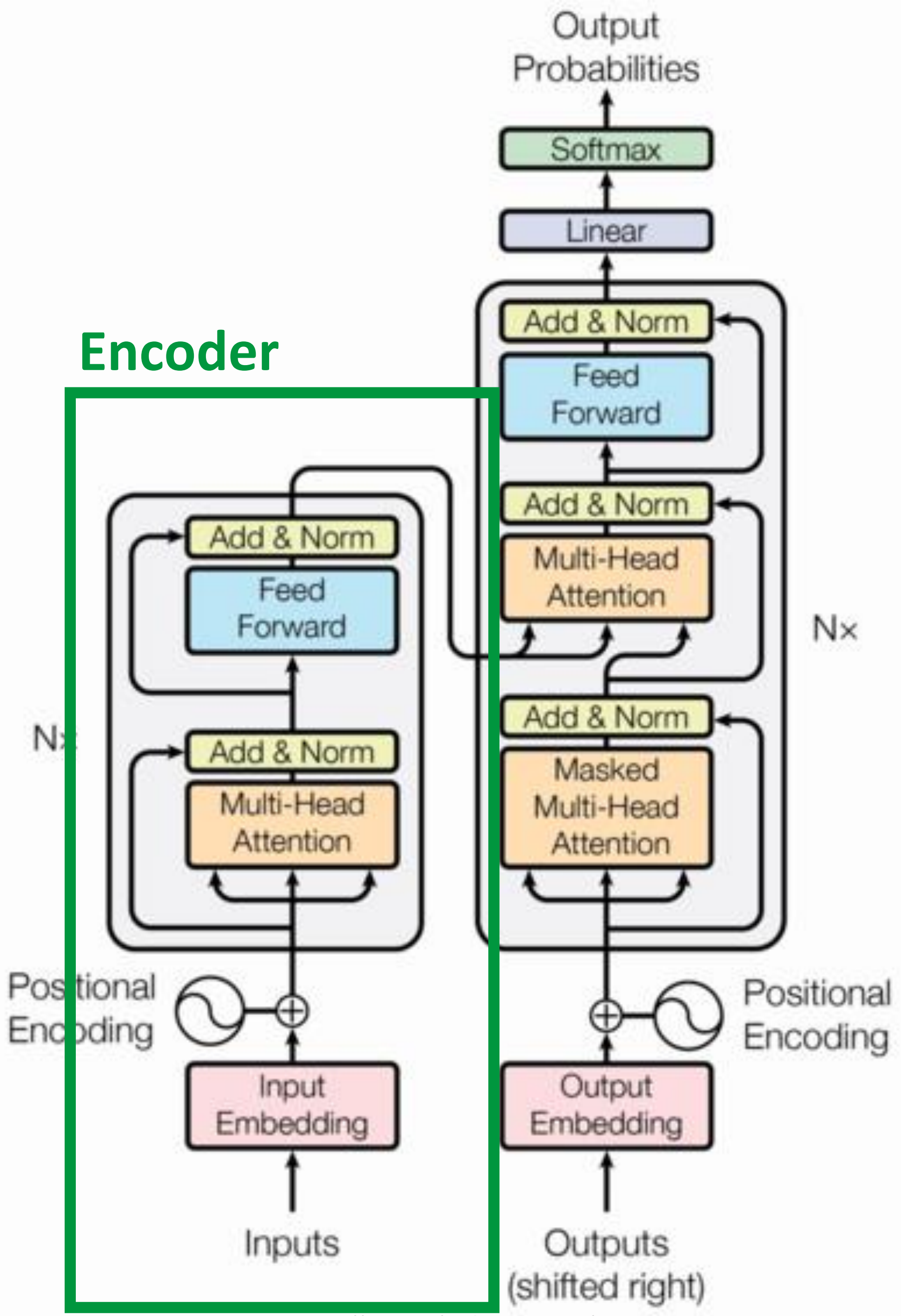

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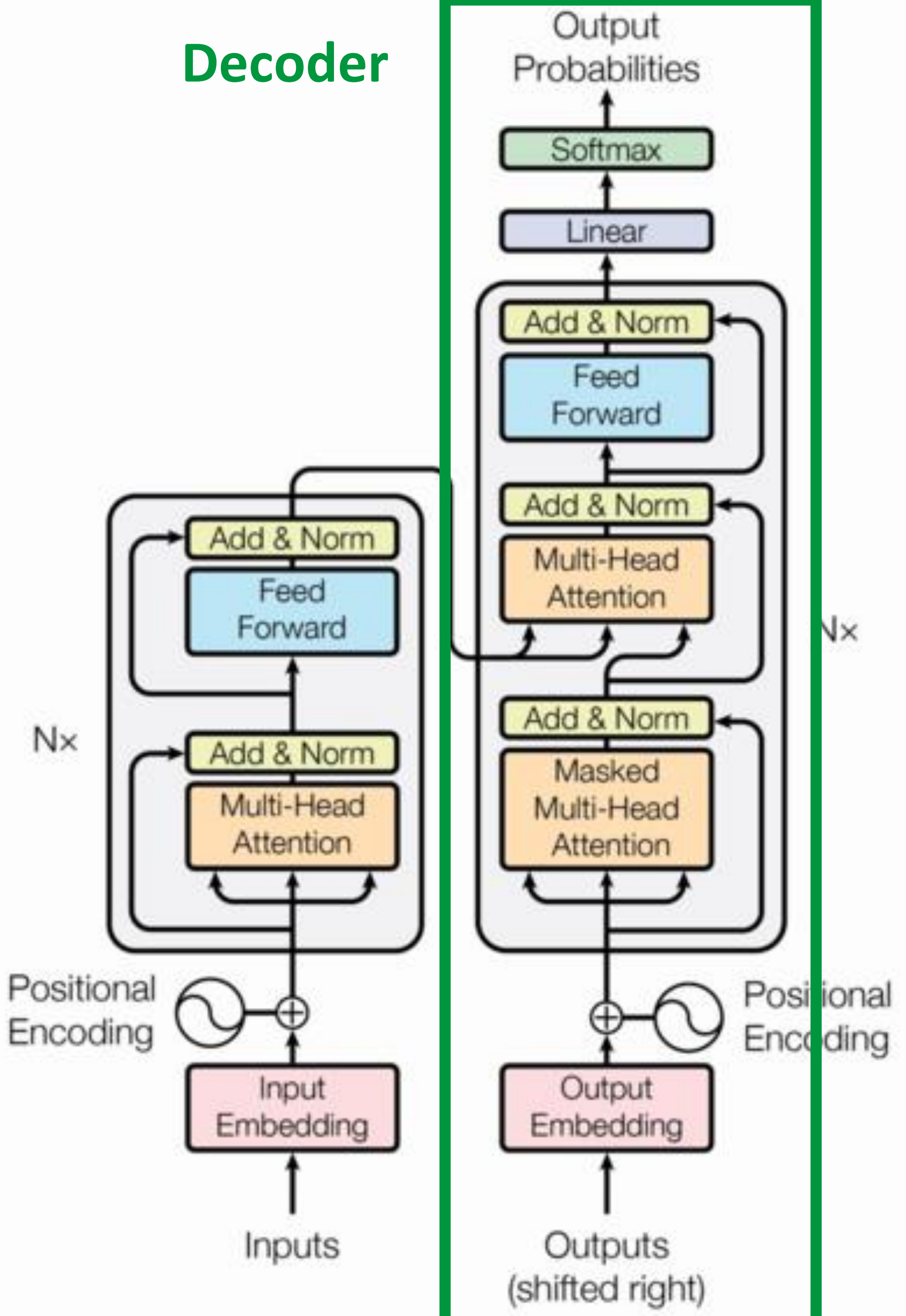

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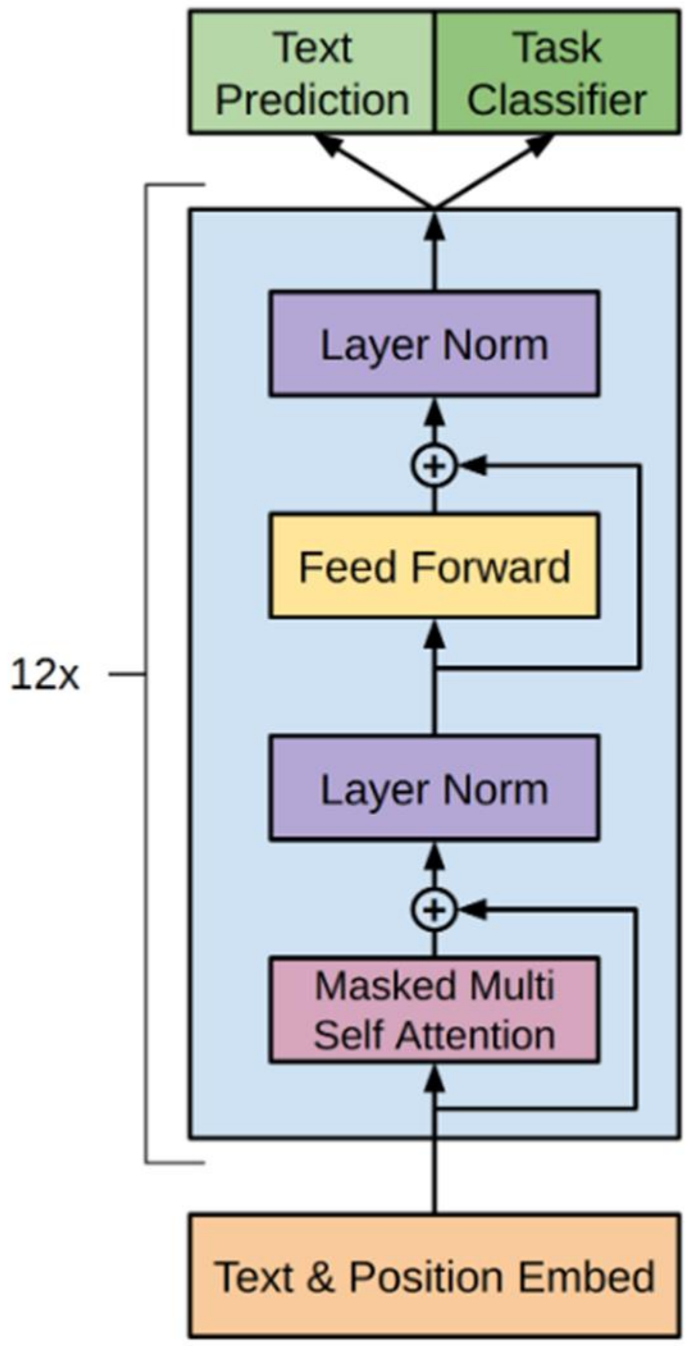

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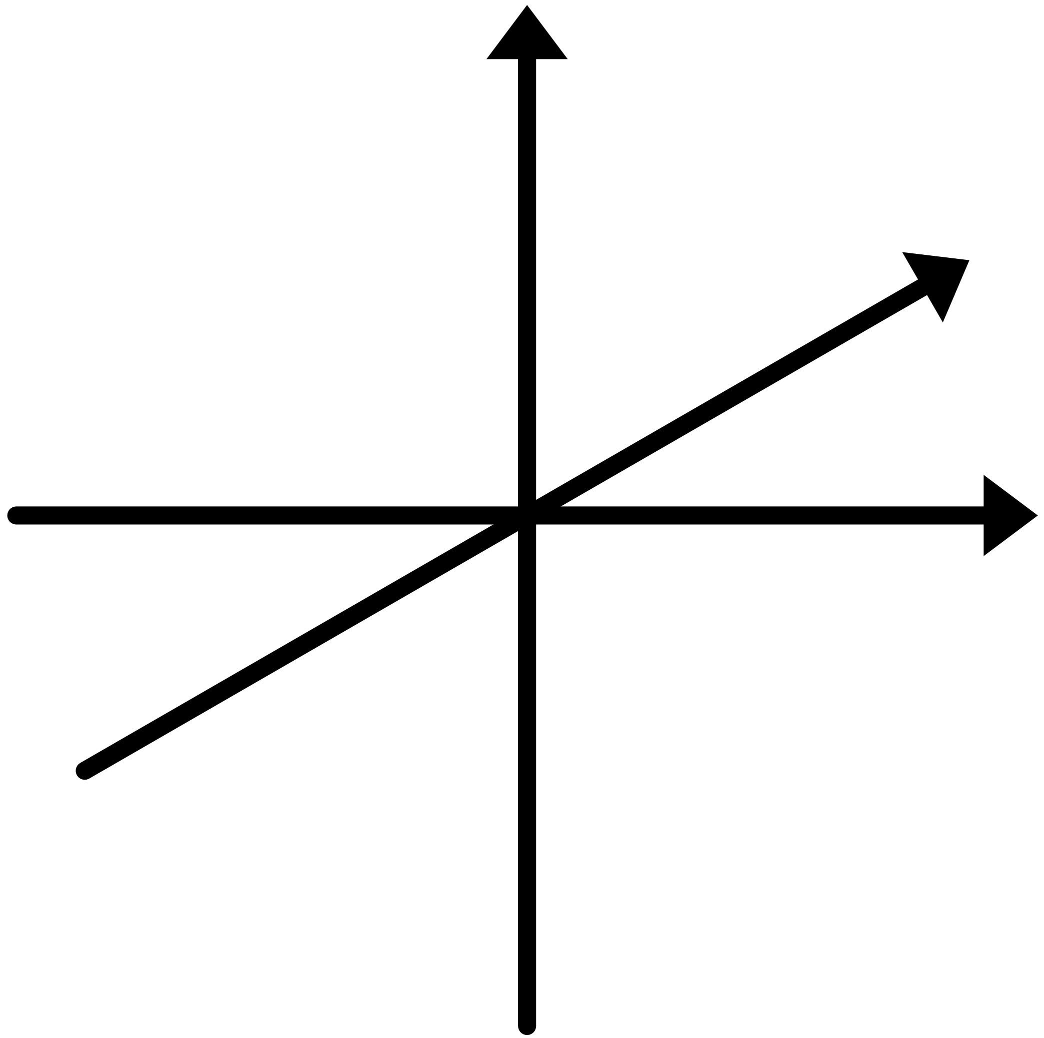
LLM Architecture

LLM Architecture

Embeddings

Goal:

- Transform words to vectors
- Maintain semantic relationships



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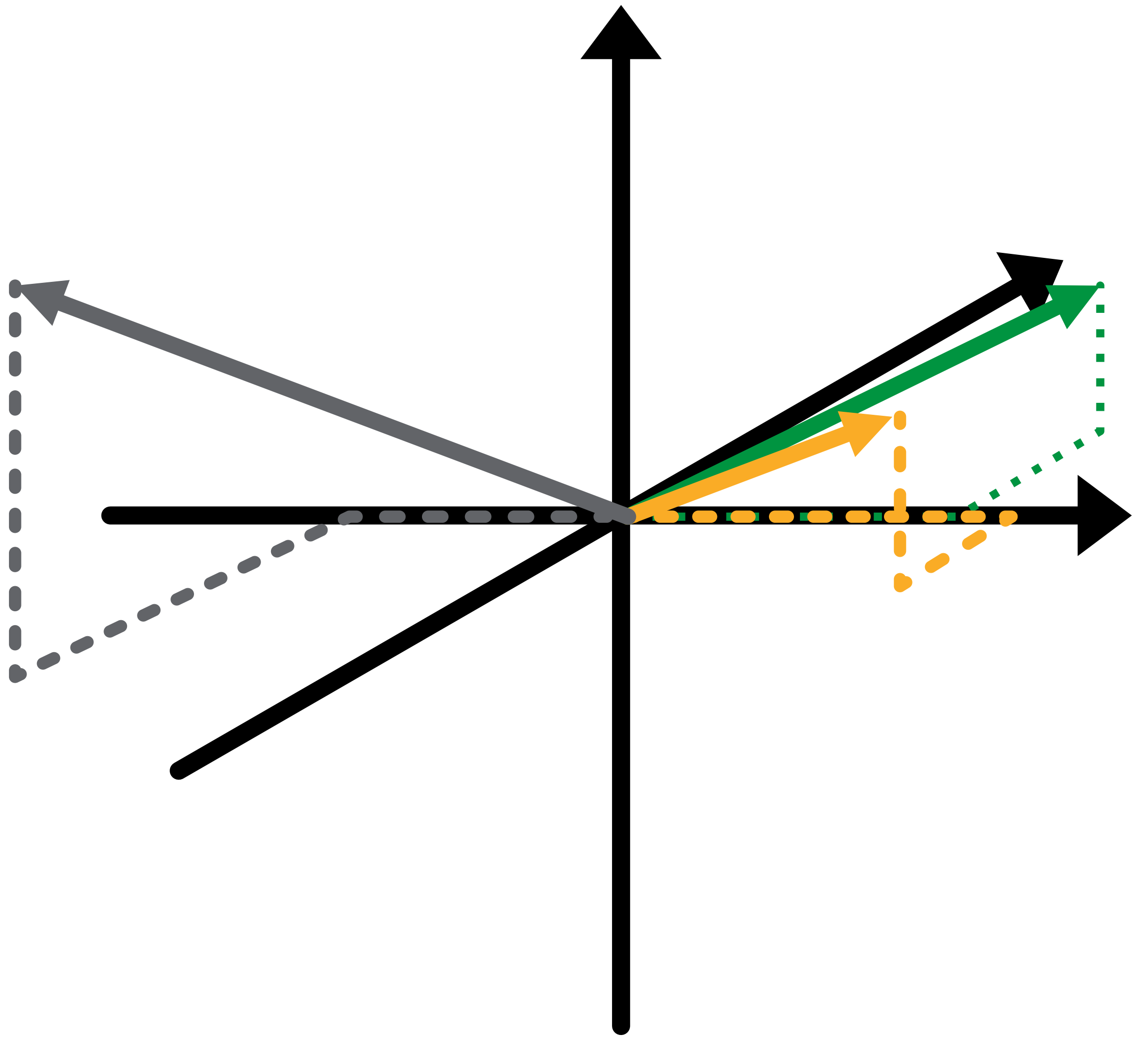

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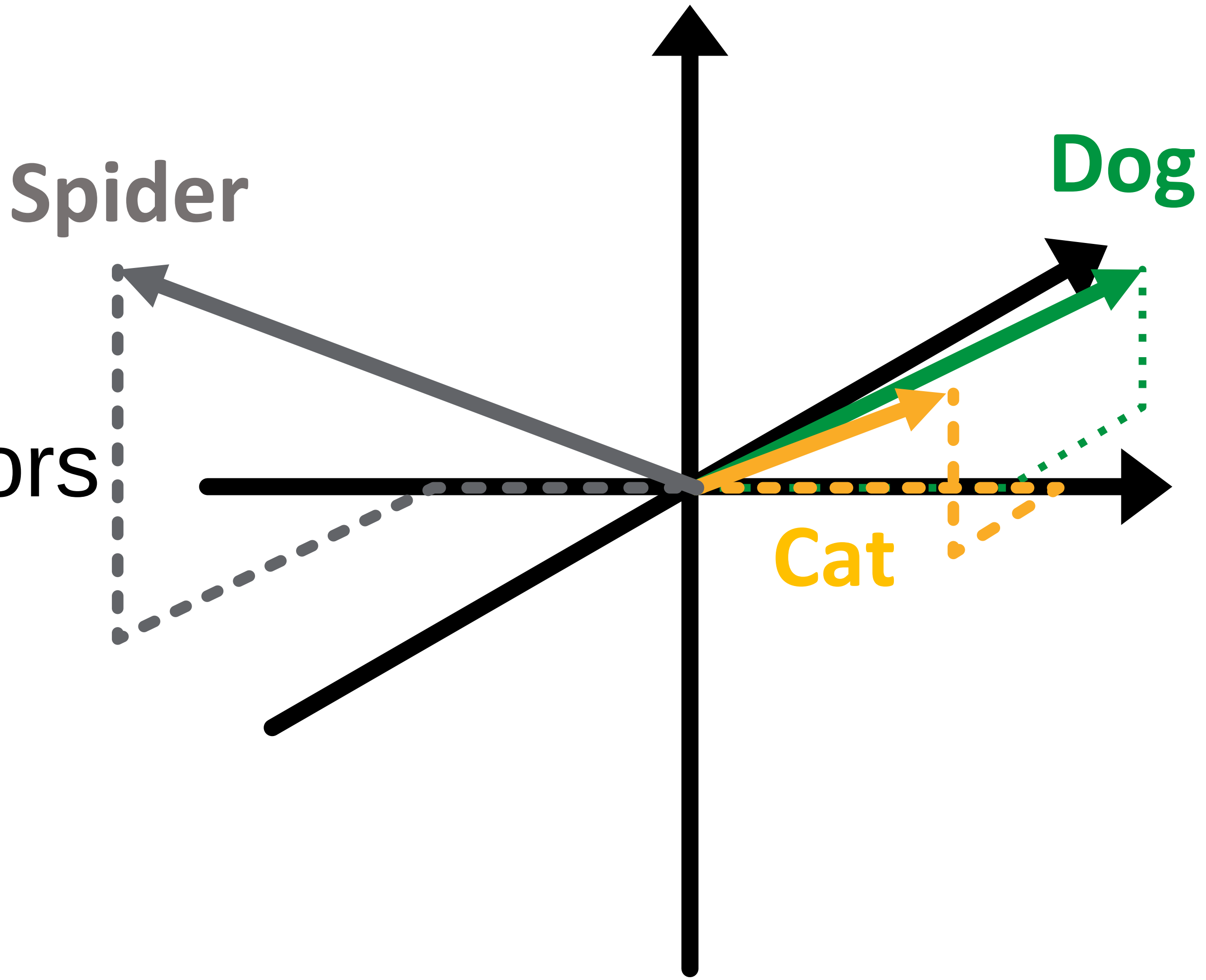

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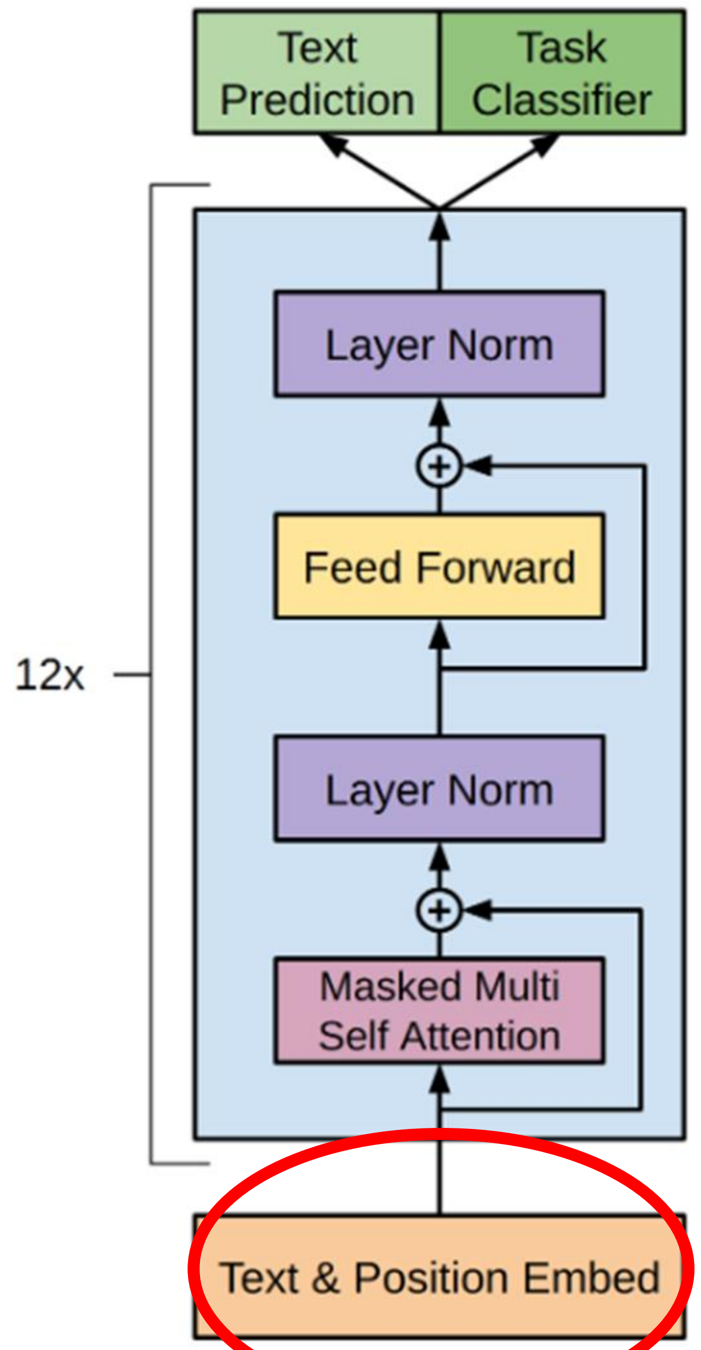

LLM Architecture

LLM Architecture

Softmax and Temperature

- Assuming we have a ML system that predicts the next word, it would give us a distribution

Softmax and Temperature

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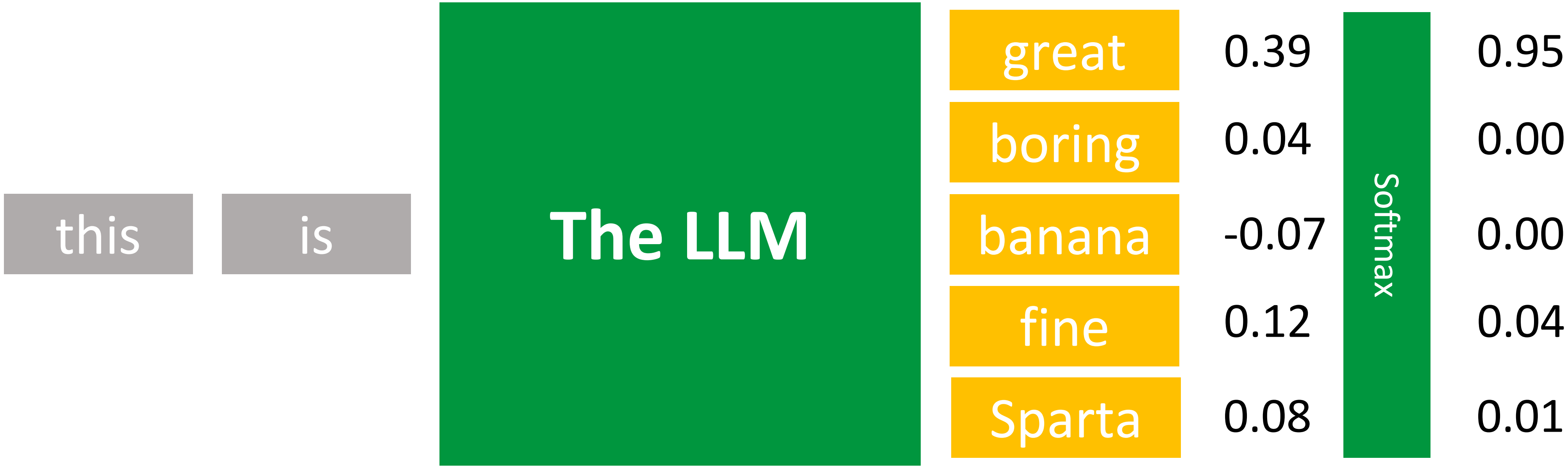

Softmax and Temperature

- Assuming we have a ML system that predicts the next word, it would give us a distribution

Sentence word

The cat slept on the ______

@NigelGebodh

https://medium.com/@nigelgebodh/why-does-my-llm-have-a-temperature-f2e314a52086

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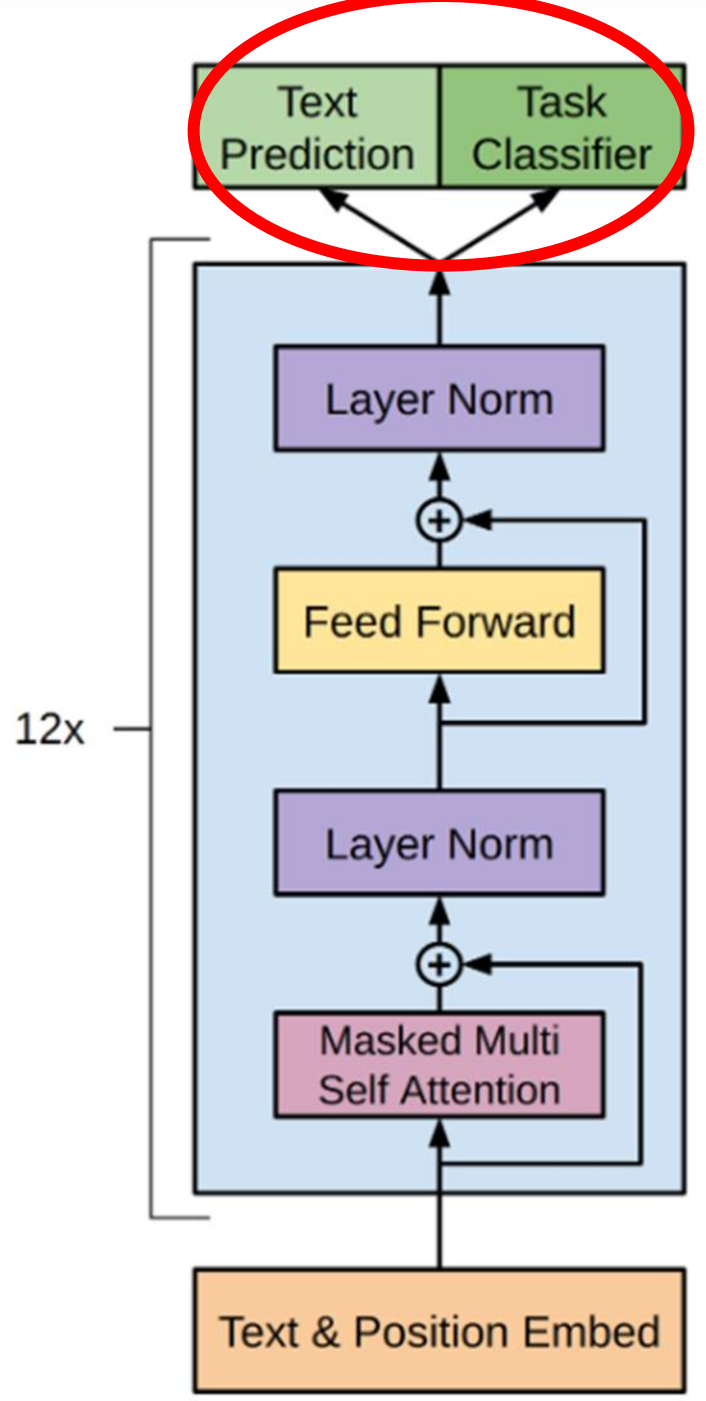

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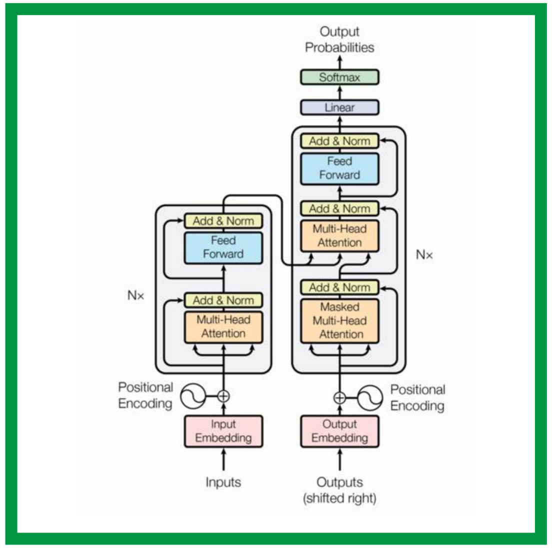
LLM Architecture

Attention is all you need Vaswani et al. 2017



model

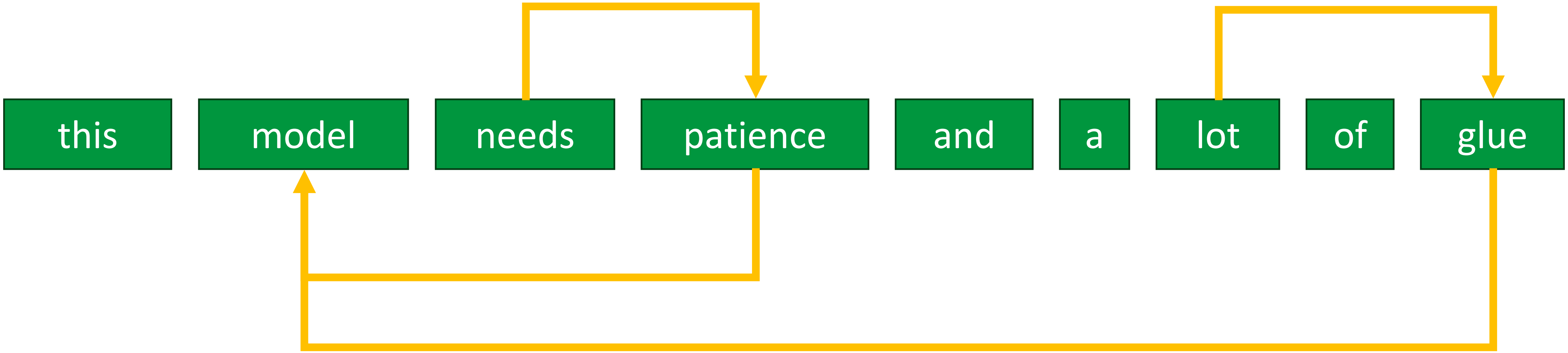
model

model

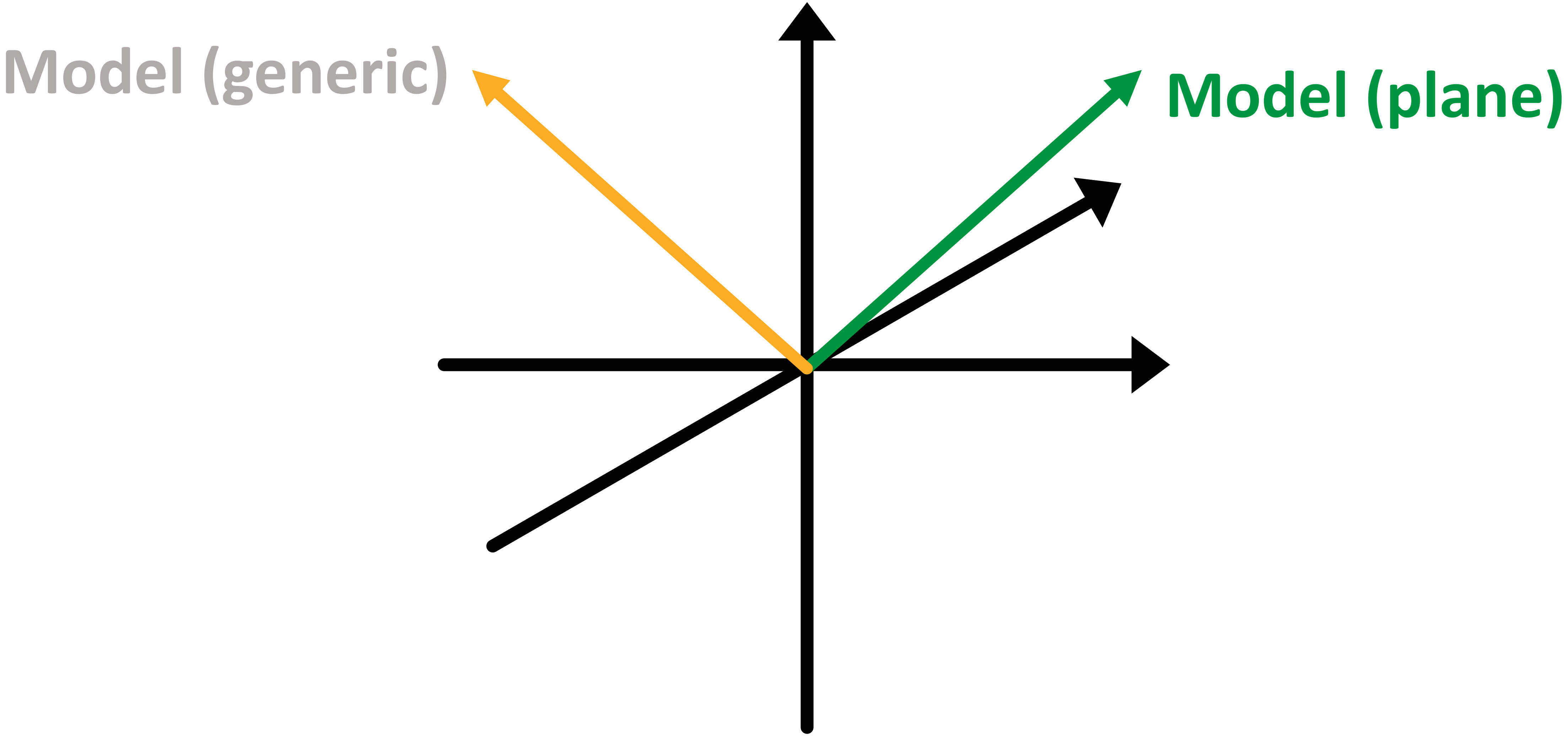
Attention Mechanism

this | model | needs | patience | and | a | lot | of | glue

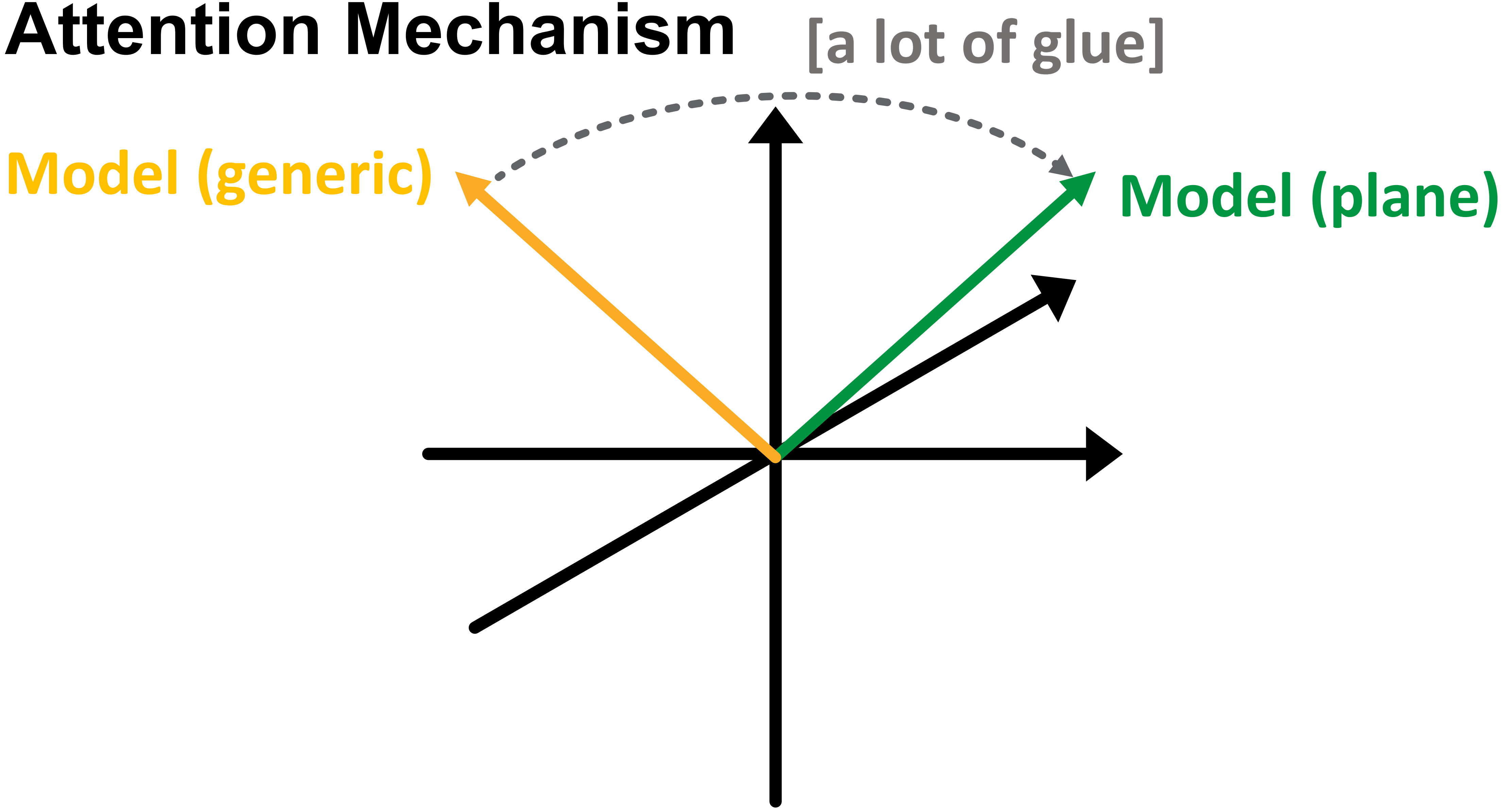
Attention Mechanism



Attention Mechanism



Attention Mechanism



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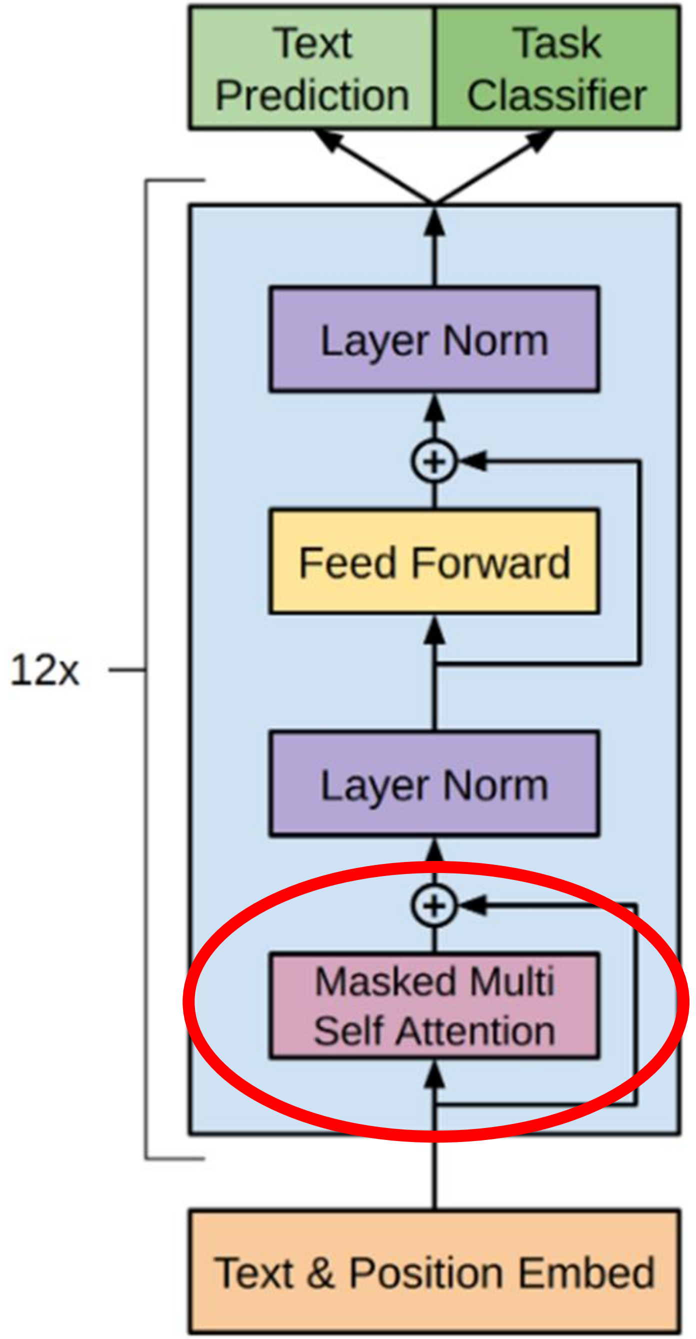

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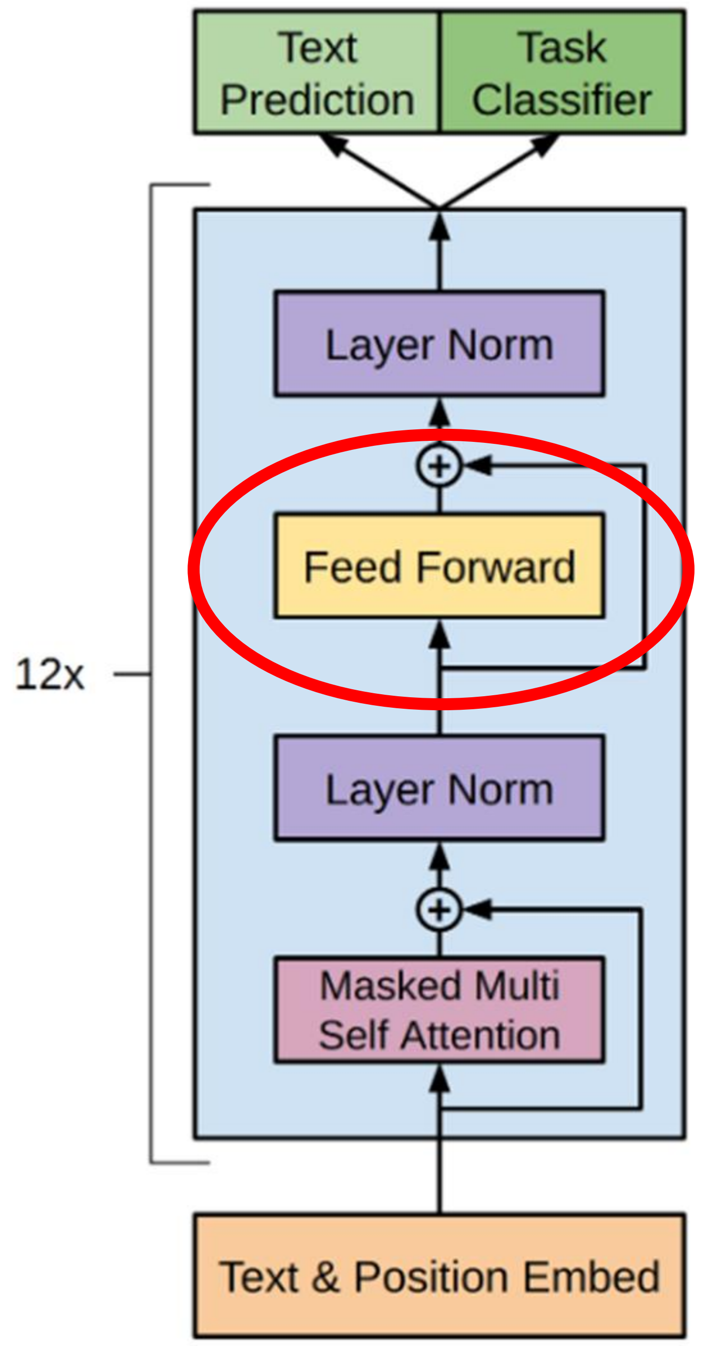

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Context

- Attention and FFwd encode context into the tokens
- The last token in the prompt (ideally) holds all the context of the prompt
- Predict the next work based on this token

- Larger attention matrices means more context

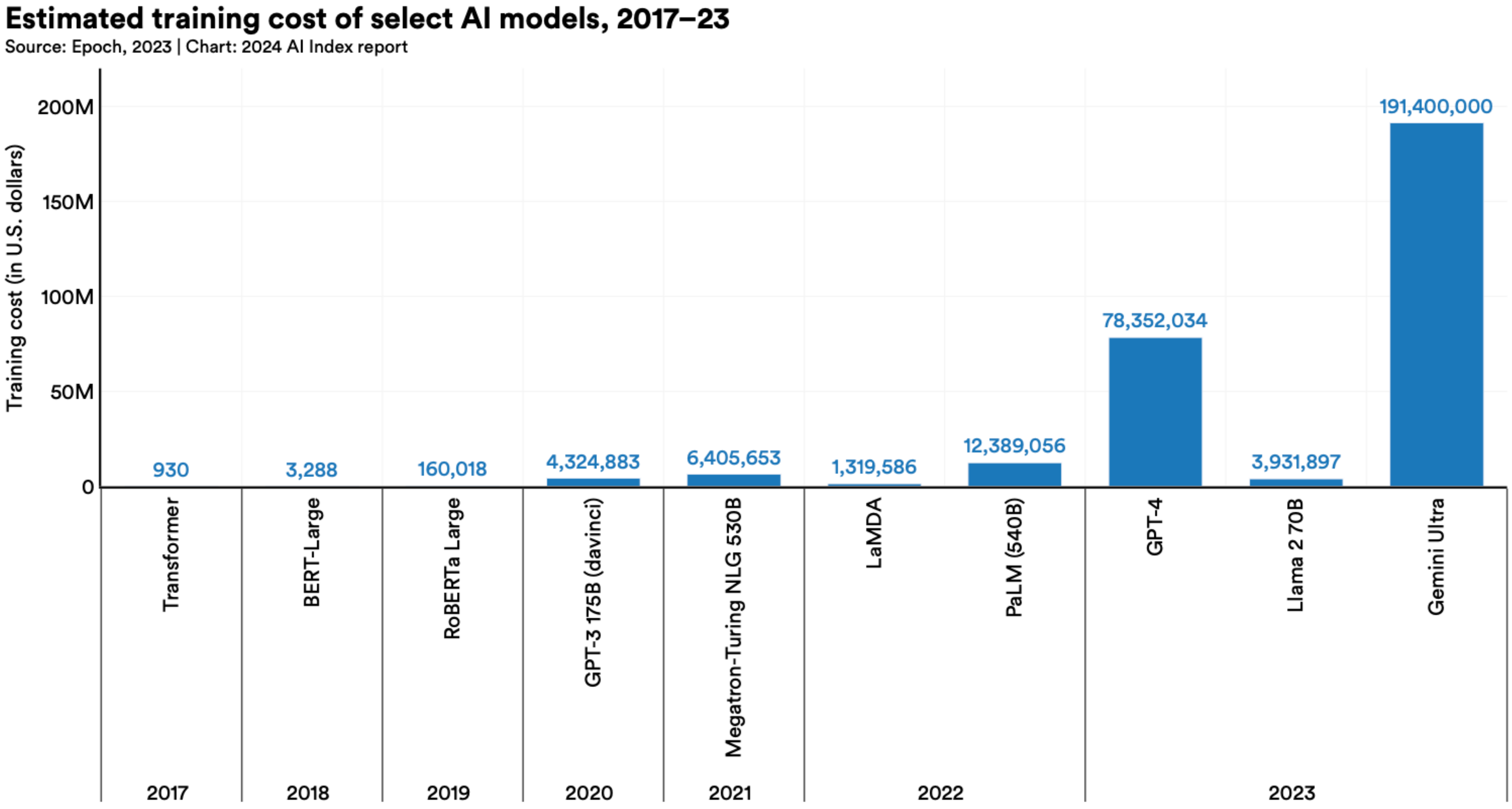
  …but also more computation

Context Length

- GPT3: 2k tokens context → 17 billion parameters
- GPT3.5: 4.5k tokens context → > 175 billion parameters*
- GPT4: 8-32k tokens context → ~1.75 trillion parameters*
- GPT4o: 128k tokens context → ???

* estimated

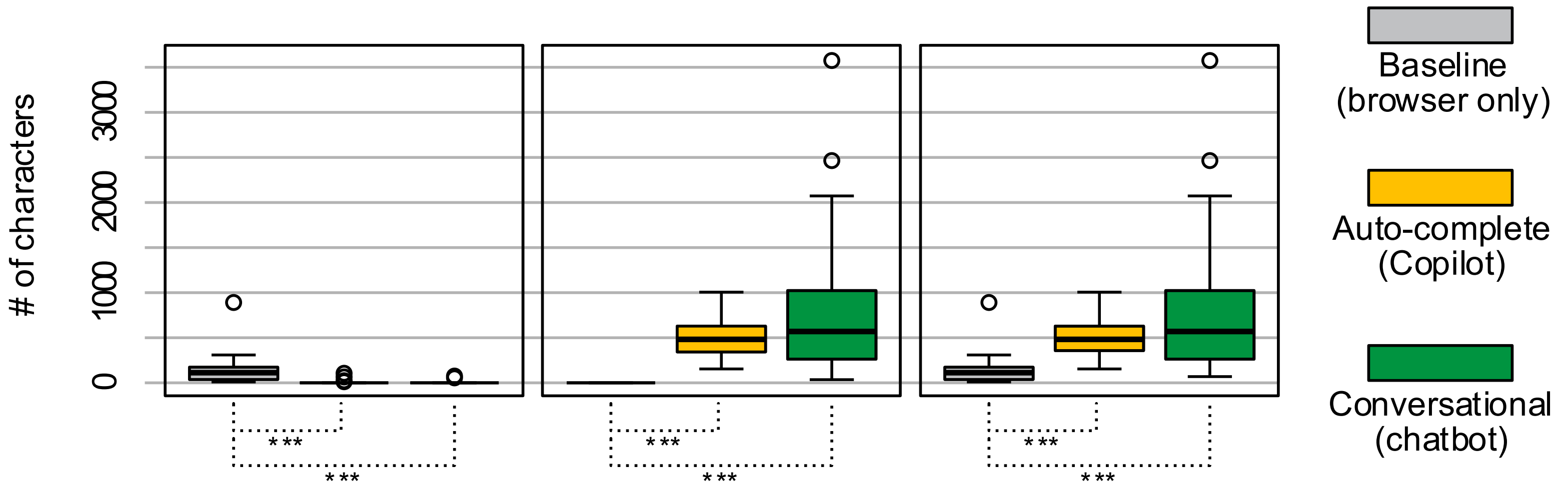
Cost of LLMs



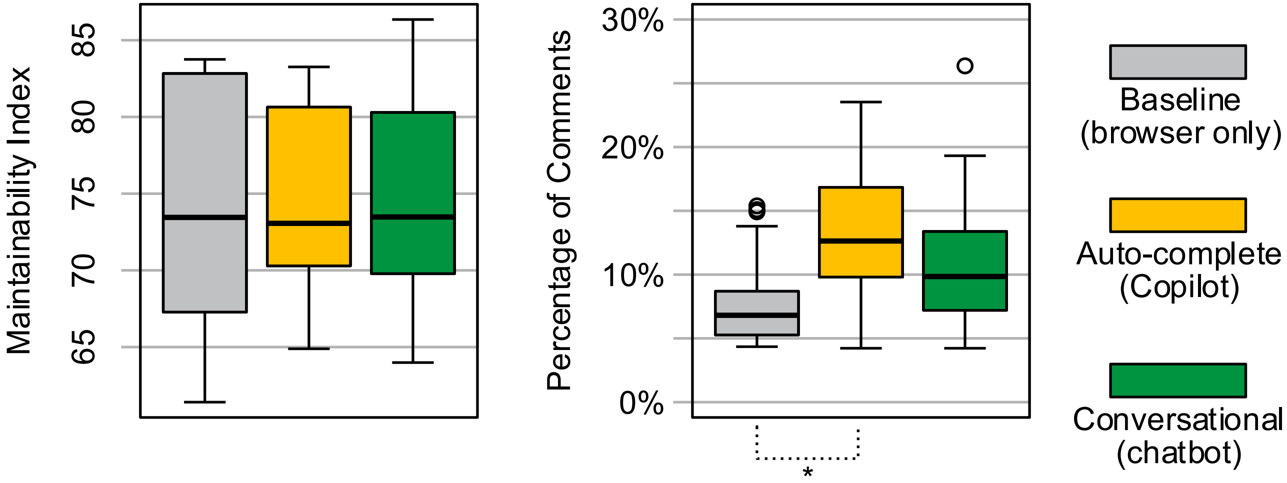
Estimated training cost of select AI models, 2017–23
Source: Epoch, 2023 | Chart: 2024 AI Index report
Training cost (in U.S. dollars)
200M
150M
100M
50M
0
930
3,288
160,018
4,324,883
6,405,653
1,319,586
12,389,056
78,352,034
3,931,897
191,400,000
Transformer
BERT-Large
RoBERTa Large
GPT-3 175B (davinci)
Megatron-Turing NLG 530B
LaMDA
PaLM (540B)
GPT-4
Llama 2 70B
Gemini Ultra
2017
2018
2019
2020
2021
2022
2023

Interacting with LLMs

Beyond Chat



# of characters
0
1000
2000
3000
***
***
***
***
***
***
Baseline (browser only)
Auto-complete (Copilot)
Conversational (chatbot)



Maintainability Index
65
70
75
80
85
Percentage of Comments
0%
10%
20%
30%
*
Baseline (browser only)
Auto-complete (Copilot)
Conversational (chatbot)

Multimodal Interaction

- Allows for interaction beyond text (e.g. video, audio, etc.)
- Typically uses other ML models to translate modality to text

https://www.youtube.com/watch?v=MirzFk_DSiI&t=196s

LLMs as system component

- API communication is often also text
- Can be processed by an LLM

- Computational overhead? Reliability?

Using LLMs locally

Vendor API vs. Local Model

Why local models?

Data
Protection

Data
Collection

Data
Introspection

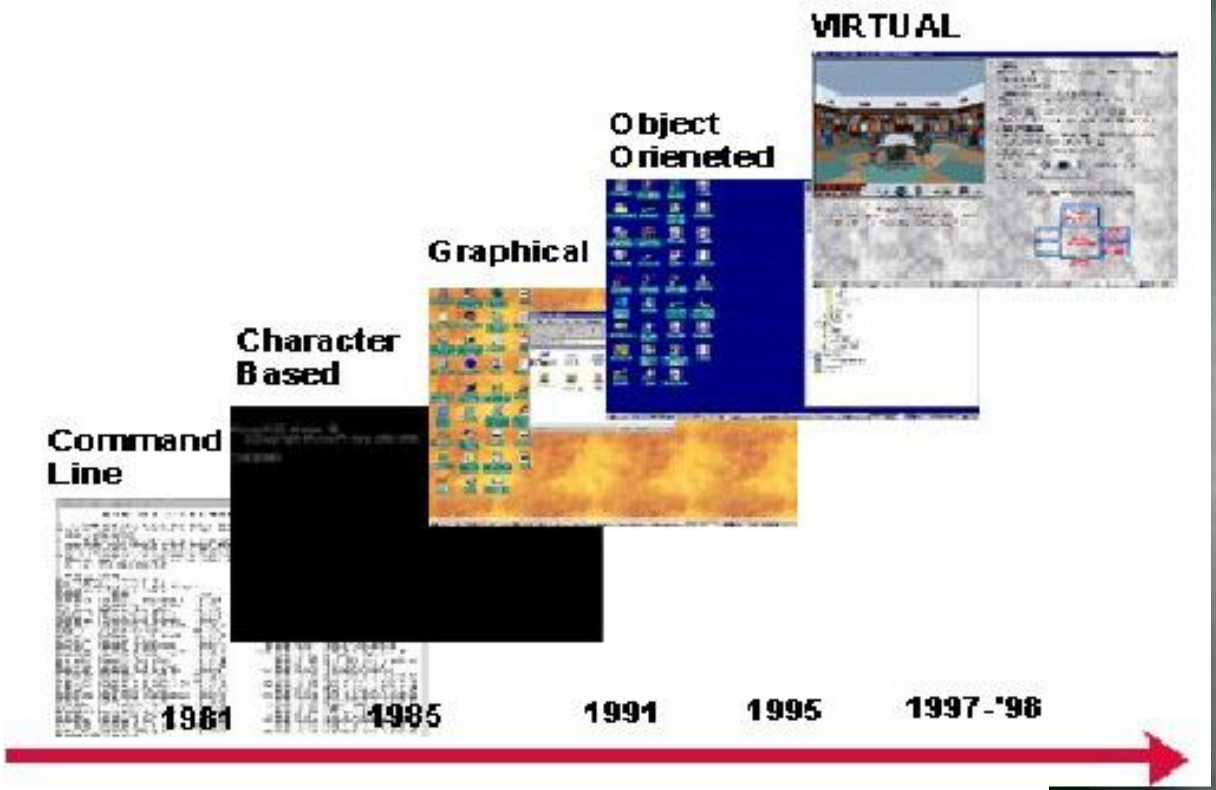
`ollama run <MODEL_NAME>`

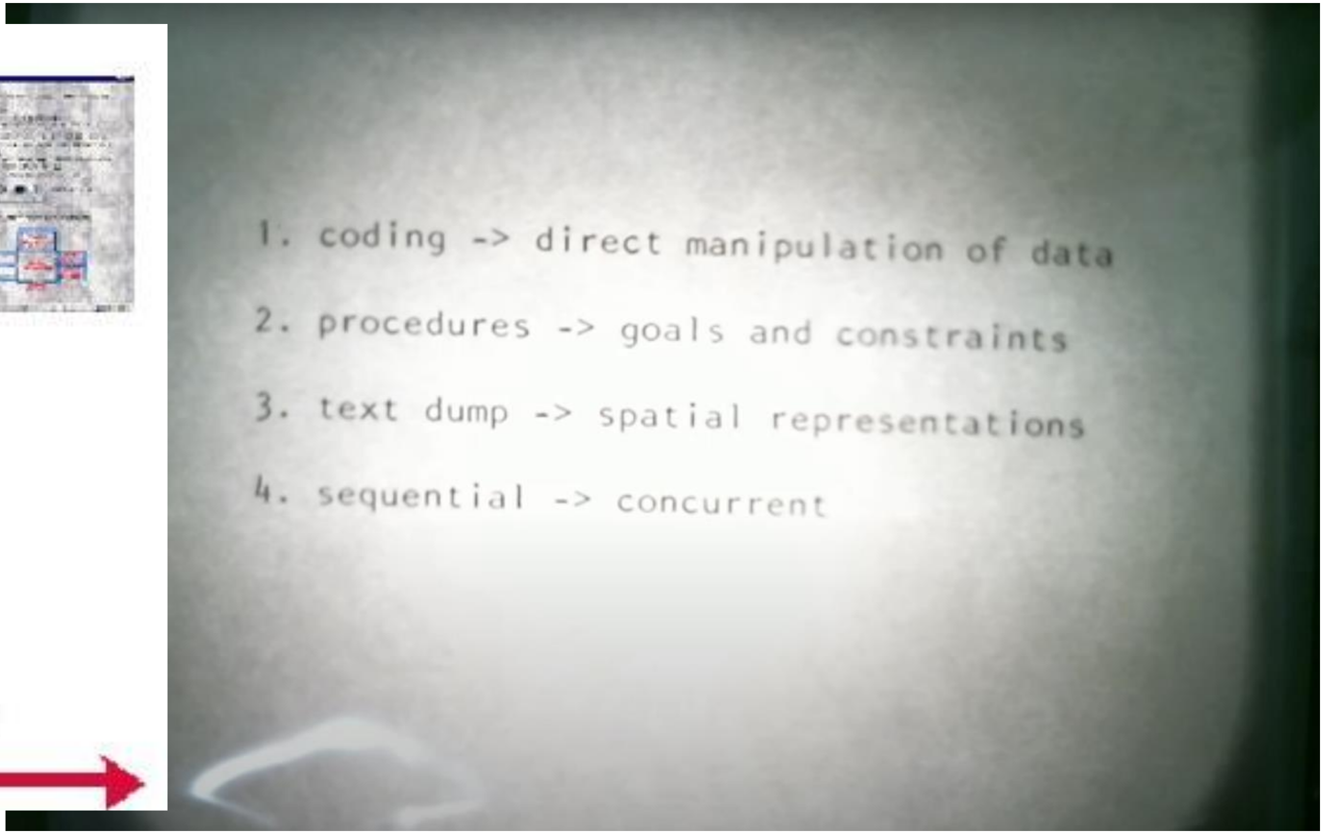
`ollama serve`

Go to `localhost:11434`

UIs for LLMs

Evolution of UIs

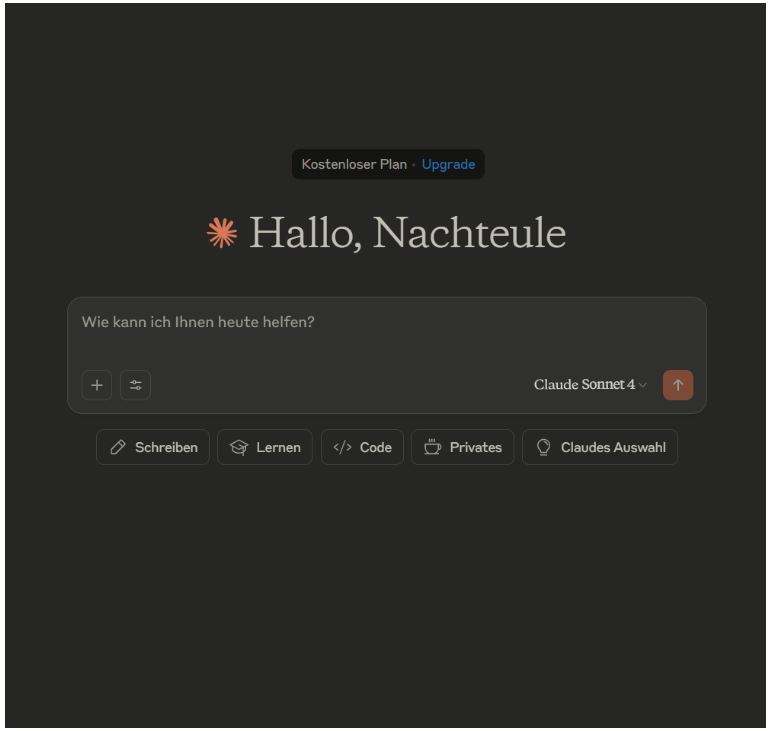


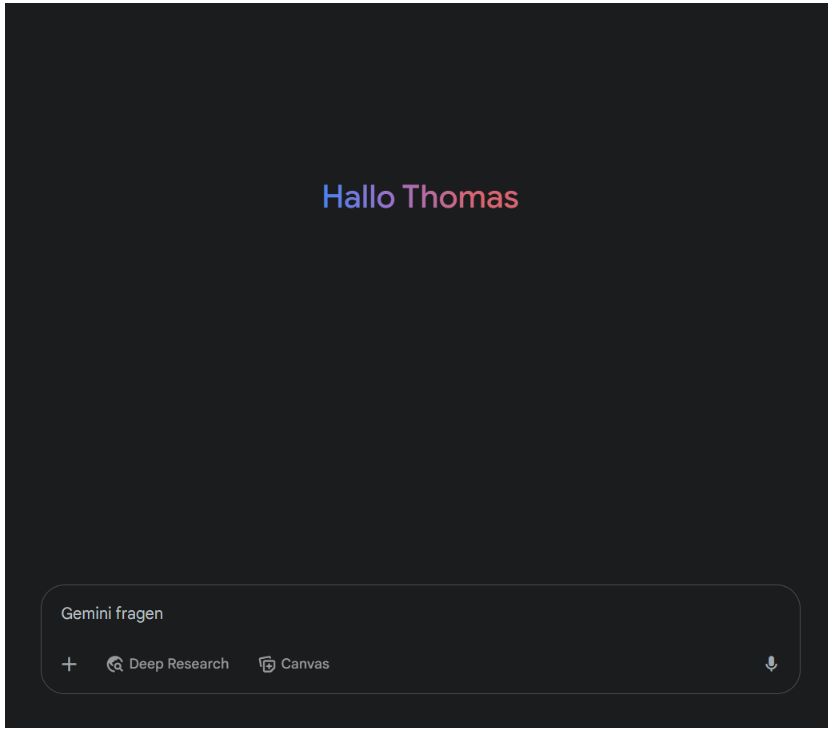


Konidaris, A., Sevasti, A., & Bouras, C. (1999). The Transition from Contemporary to Virtual User Interfaces for Web-Based Services. In *WebNet World Conference on the WWW and Internet* (pp. 125-130). Association for the Advancement of Computing in Education (AACE).

Bret Victor – The Future of Programming

The Evolution of UIs





Questions?

Contact

Thomas Weber
LMU Munich
thomas.weber@ifi.lmu.de